Gemshorn Build Methods And Slip-Cast Family Print Packet

Generated: 2026-05-06

Packet folder: `/mnt/c/Users/Tony/Documents/GitHub/gemshorn`

File Map

File	Purpose
`design.md`	Project intent, catalog metadata, assumptions, and validation plan.
`bom.csv`	Starter bill of materials with part categories, quantities, drawing refs, and notes.
`sourcing.csv`	Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.
`cut-list.csv`	Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.
`drawing-brief.md`	Manufacturing drawing and technical product sketch brief.
`assembly-manual.md`	Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.
`validation.csv`	Target/measured values, tolerance, environment, result, and tuning/build action log.
`supplier-rfq.md`	Supplier email/request-for-quote starter.
`visual-bom-brief.md`	Art direction for an image-forward visual BOM.
`wolfram-starter.wl`	Wolfram starter for physics, optimization, visualization, and validation.
`README.md`	Project artifact.
`authentic-horn-build-plan.md`	Project artifact.
`authenticity-notes.md`	Project artifact.
`build-methods.md`	Project artifact.
`family-spec.csv`	Project artifact.
`hole-schedule-historical.csv`	Project artifact.
`hole-schedule-modern.csv`	Project artifact.
`horn-blank-spec.csv`	Project artifact.
`material-options.md`	Project artifact.
`mold-and-slip-casting-plan.md`	Project artifact.
`photo-shotlist.md`	Project artifact.
`risks.md`	Project artifact.
`sources.md`	Project artifact.
`tuning-and-fingering.md`	Project artifact.
`validation-report.md`	Project artifact.

design.md

Project intent, catalog metadata, assumptions, and validation plan.

Slip-Cast Gemshorn Family Build Packet

Generated: 2026-05-02

v4 refresh: 2026-05-06

Packet root: `/mnt/c/Users/Tony/Documents/GitHub/gemshorn`

Scope

This packet gives you a historically informed gemshorn build path and a practical slip-cast consort family. The two are intentionally separated:

- **Historical archetype:** short curved horn or clay horn, open wide end closed with a fipple/block, closed pointed tip, limited holes, limited range. This is the authenticity target.
- **Slip-cast family:** repeatable ceramic bodies in F/C consort sizes with seven front holes and one thumb/tuning vent. This is the practical ensemble and production target, and it is marked as a modern reconstruction approach.

Deliverables

- `gemshorn-design-table.xlsx` - parametric workbook with inputs, family dimensions, hole schedules, mold scaling, and validation targets.
- `family-spec.csv` - fired and master dimensions for the slip-cast family.
- `hole-schedule-modern.csv` - modern seven-front-plus-thumb hole diameters and positions.
- `hole-schedule-historical.csv` - historically informed four-front-plus-thumb pilot schedule.
- `drawings/` - SVG shop drawings for body section, family scale, hole layout, voicing detail, mold schematic, and visual BOM.
- `cad/gemshorn_family.scad` - OpenSCAD starter for master geometry.
- `cnc/mold-master-cam-plan.md` - CAM/print/fixture planning notes.
- `bom.csv`, `sourcing.csv`, `cut-list.csv`, `validation.csv` - production and validation tables.
- `assembly-manual.md`, `mold-and-slip-casting-plan.md`, `tuning-and-fingering.md`, `authenticity-notes.md` - shop-facing instructions.
- `authentic-horn-build-plan.md`, `horn-blank-spec.csv` - natural-horn authentic build path and blank selection specs.
- `build-methods.md` - side-by-side real horn, slip-cast ceramic, CNC split wood, and research-material build methods.
- `material-options.md` - candidate materials and safety/finish gates.
- `risks.md`, `photo-shotlist.md`, `site/index.html` - v4 risk, photo, and build-log-site deliverables.
- `cnc/cnc-plan.json`, `cnc/operations.csv`, `cnc/setup-sheet.md`, `cnc/wood-body-cnc-plan.md` - CNC/router planning deliverables.
- `wolfram/instrument-model.wl` - v4.2 Wolfram package.
- `wolfram-starter.wl` - notebook-ready physics starter.

Acoustic Model

The working model is a Helmholtz/vessel-flute model, not an open-open flute model:

```
f = c/(2*pi) * sqrt(G_total / V)
G_total = sum(A_i / L_eff_i)
L_eff circular hole approx wall + 0.85*r
```

For a tuning hole with required incremental conductance `G`, the first-pass hole radius is:

```
r = (0.85*G + sqrt((0.85*G)^2 + 4*pi*G*wall)) / (2*pi)
```

The body dimensions use geometric scaling from the soprano C5 prototype. That keeps volume, window area, hole area, and neck length in the same acoustic family. Final pitch still requires empirical tuning because ceramic shrinkage, glaze, fipple behavior, hole chamfer, and actual cavity volume dominate the last cents.

Slip-Cast Family

ID	Key	Hz	Fired L in	Master L in	Wide OD in	Volume ml	Window W in	Window H in
GEM-SC-F3	F3	174.614	27.719	31.499	4.645	2401.814	1.289	0.420
GEM-SC-C4	C4	261.626	18.500	21.023	3.100	649.562	0.860	0.280

GEM-SC-F4	F4	349.228	13.859	15.749	2.322	260.245	0.644	0.210
GEM-SC-C5	C5	523.251	9.250	10.511	1.550	71.554	0.430	0.140
GEM-SC-F5	F5	698.456	6.930	7.875	1.161	27.799	0.322	0.105

Historical Archetype

Prototype: `GEM-HIST-G4` in `G4`, fired centerline length `12.35 in`, wide OD `2.07 in`, volume `180 ml`.

The historical model should be built first with waxable/reworkable holes. If you want the strictest Virdung/Agricola reading, make three front holes plus a back/thumb hole, then treat the fourth front hole in `hole-schedule-historical.csv` as an optional clay-find/modern-convenience variant.

Manufacturing Assumptions

- Fired clay shrinkage: `12.0%` linear. Update this from your actual clay body test bars before committing to molds.
- Clay body: white stoneware or porcelain casting slip, cone 5/6 or cone 10 depending on shop practice.
- Wall: scaled from `0.130 in` at soprano C, clamped between `0.110 in` and `0.240 in`.
- Holes: cast-in undersize dimples or drill leather-hard to the greenware start diameters, then tune by opening slowly after bisque.
- Finish: exterior glaze only around body; keep windway, labium edge, tone-hole interiors, and seating surfaces unglazed or very lightly treated.

Build Methods

Method	Status	Primary Files	Notes
Natural horn	Historically informed reference	`authentic-horn-build-plan.md`, `horn-blank-spec.csv`	Tune each horn as a one-off; no endangered/wild horn sourcing.
Slip-cast ceramic	Production family path	`mold-and-slip-casting-plan.md`, `family-spec.csv`, `hole-schedule-modern.csv`	Best repeatability after clay shrinkage is measured.
CNC-routed split wood	Modern prototype path	`cnc/wood-body-cnc-plan.md`, `cad/gemshorn_split_wood_body.scad`, `drawings/gemshorn-cnc-wood-body.svg`	Good for fast fipple, volume, and hand-layout tests.
Alternate materials	Research path	`material-options.md`, `risks.md`, `validation.csv`	Must pass mouth-safety, edge-retention, and leak tests.

Hardware Alignment

Operation	Tool / Fixture	Applies To	Release Check
Horn blank fitting	Fine saw, scraper, drill press, tapered reamers	Natural horn	Tip remains closed; fipple block seats without leaks.
Mold master shaping	3D print, CNC tooling board, or sealed master	Slip-cast ceramic	Master is scaled by measured shrinkage and releases from plaster.
Plaster mold making	Two-part mold box, registration keys, pour mouth	Slip-cast ceramic	Mold halves register and cast releases at leather hard.
Split-wood CNC routing	1/4 in upcut, 1/4 in ball end, 1/8 in ball end, dowel-pin fixture	CNC wood body	Halves register, cavity leak test passes before shaping.
Hole tuning	Pin gauges, drill bits, tapered reamers, wax	All methods	Holes start undersize and tune low to high.
Validation	Tuner, thermometer, water-fill volume setup, calipers	All methods	`validation.csv` records measured values, cents error, and action.

Validation Gates

- Measure fired cavity volume by water fill before final tuning.
- Record closed note, all scale degrees, and high/tuning vent at 68 F reference or with temperature logged.
- Tune by opening holes only. Lower pitch by wax, removable clay-safe filler, or ring/vent control.
- Reject or rework bodies with warped windways, rounded labium edges, cracks at holes, or shrinkage outside the measured clay-body range.

Open Decisions

- Actual clay body, firing cone, glaze schedule, and shrinkage test result.
- Whether the production family should be C/F consort, G/D historical consort, or both.
- Whether the tuning ring is a brass sleeve, leather wrap with rotating vent, or ceramic collar.
- Whether CNC wood bodies should use a permanent glue seam, a gasketed service seam, or both as parallel prototypes.
- Which alternate material coupons are worth testing after ceramic/wood/horn baselines.

bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

item	part	qty	material_spec	make_buy	drawing_ref	notes
BOM-000	Natural horn blank	3 blanks for first authentic prototype	Domesticated cow/ox/goat/sheep horn, closed tip, no cracks	buy	horn-blank-spec.csv	Authentic build path; avoid endangered/wild chamois sourcing
BOM-001	Slip-cast ceramic body	1 per instrument	White stoneware or porcelain casting slip, measured shrinkage	make	drawings/gemshorn-body-section.svg	Use family-spec.csv for size-specific fired and master dimensions
BOM-002	Two-part plaster mold	1 per size	Pottery plaster or equivalent	make	drawings/gemshorn-mold-schematic.svg	Add registration keys and pour mouth
BOM-003	Master pattern	1 per size	3D printed resin/PLA, tooling board, or sealed wood	make or outsource	cad/gemshorn_family.scad	Scale for shrinkage and seal before plaster
BOM-004	Fipple/block	1 per instrument	Cedar, maple, pear, plaster, or stable hardwood	make	drawings/gemshorn-voicing-detail.svg	Removable block recommended for tuning and service
BOM-005	Seal wrap/gasket	1 set per instrument	Leather, cork, beeswax, or thin silicone gasket	buy	drawings/gemshorn-voicing-detail.svg	Keep mouth-contact material safe and replaceable
BOM-006	Optional tuning ring/collar	1 per instrument	Thin brass/stainless ring, leather band, or ceramic collar	buy or make	drawings/gemshorn-voicing-detail.svg	Rotates over vent to lower closed keynote
BOM-007	Tuning and measurement kit	1 shop set	Pin gauges, drill bits, reamers, tuner, thermometer, wax	buy	validation.csv	Required before final pitch claims
BOM-008	CNC split-wood body blanks	2 halves per prototype	Maple, cherry, pear, beech, or boxwood; straight-grained; oversized for routing	make	drawings/gemshorn-cnc-wood-body.svg	Modern CNC router build path; start with C5 or G4
BOM-009	Wood body registration dowels	2 to 4 per prototype	1/4 in hardwood dowels or metal datum pins	buy	cnc/wood-body-cnc-plan.md	Outside final body outline or hidden after shaping
BOM-010	Wood body adhesive/seal system	1 set per prototype	Thin even glue line plus mouth-safe interior/finish test coupon	buy	risks.md	Critical leak and mouth-safety gate
BOM-011	Alternate material coupons	1 set	Stoneware, porcelain, maple/cherry/pear, printed master material, optional acrylic teaching coupon	make or buy	material-options.md	Use coupons before committing alternate materials to instruments

sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

item	spec	supplier_or_search_terms	price_each	date_checked	lead_time	substitution_rule	risk
Natural horn blank	Domesticated cow/ox/goat/sheep horn, closed tip, cleanable interior	cow horn blank ox horn craft horn goat horn instrument blank	TBD live check		TBD	Do not use endangered species; reject cracked or delaminated blanks	Each horn has different volume and curvature, so every blank is tuned empirically
Casting slip	White stoneware or porcelain casting slip; shrinkage data required	ceramic casting slip cone 6 porcelain stoneware gallon	TBD live check		TBD	Must provide shrinkage and firing cone	Shrinkage variation changes all dimensions and pitch
Pottery plaster	No. 1 pottery plaster or equivalent mold plaster	pottery plaster mold plaster 50 lb	TBD live check		TBD	Use ceramic mold plaster, not construction plaster	Poor absorption causes slow wall growth and release trouble
Hardwood block stock	Fine-grain stable hardwood blanks, 1/2 to 2 in depending size	cedar pear maple turning blank instrument block	TBD live check		TBD	Avoid resinous or mouth-unsafe finishes	Block swelling or warping changes windway
Leather/cork gasket	Thin natural leather, cork sheet, or mouth-safe gasket material	thin cork gasket sheet leather scrap beeswax	TBD live check		TBD	Must seal without shedding dust	Leaks make voicing unstable
Tuning ring material	Thin brass strip/tube or leather collar	thin brass strip tube hobby K&S brass	TBD live check		TBD	Collar must rotate smoothly and not chip clay	Rough collar damages vent edge
CNC wood body blank	Straight-grained maple, cherry, pear, beech, or boxwood; two oversized halves per prototype	hardwood turning blank maple cherry pear boxwood 12 inch	TBD live check		TBD	Substitute only with stable, mouth-safe-finish-compatible hardwood	Glue seam and humidity movement can shift tuning
Registration dowels	1/4 in hardwood dowels or reusable metal datum pins	quarter inch hardwood dowel precision dowel pins	TBD live check		TBD	Pin diameter must match CAM and drill operation	Registration error ruins split-body alignment
Mouth-safe finish test materials	Shellac flakes, food-safe oil/wax, or proven instrument finish for coupons	shellac flakes food safe wood finish beeswax carnauba	TBD live check		TBD	Must pass mouth-contact and odor test before use	Finish can swell/round windway and labium
Prototype/master print material	PLA/PETG/SLA resin for sealed masters only unless mouth safety is proven	3D print filament resin mold master sealant	TBD live check		TBD	Final playing body requires safety review	Layer lines leak and roughen windway

cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

part_id	part	qty	rough_size	finished_size	material	operation	yield_or_offcut	notes
GEM-SC-F3-MASTER	Bass F scaled master	1	33.50 x 7.28 x 7.28 in envelope	31.50 in centerline, 5.28 in wide OD	sealed print/tooling board	print/CNC, seal, sand, polish, mark dimples	one master per size	Check mold release before committing plaster
GEM-SC-F3-MOLD	Bass F plaster mold	1	mold box about 35.50 x 9.28 x 9.28 in	two plaster halves plus pour opening	pottery plaster	cast mold halves around sealed master	reuse for production run after seasoning	Add registration keys and split-line witness marks
GEM-SC-C4-MASTER	Tenor C scaled master	1	23.02 x 5.52 x 5.52 in envelope	21.02 in centerline, 3.52 in wide OD	sealed print/tooling board	print/CNC, seal, sand, polish, mark dimples	one master per size	Check mold release before committing plaster
GEM-SC-C4-MOLD	Tenor C plaster mold	1	mold box about 25.02 x 7.52 x 7.52 in	two plaster halves plus pour opening	pottery plaster	cast mold halves around sealed master	reuse for production run after seasoning	Add registration keys and split-line witness marks
GEM-SC-F4-MASTER	Alto F scaled master	1	17.75 x 4.64 x 4.64 in envelope	15.75 in centerline, 2.64 in wide OD	sealed print/tooling board	print/CNC, seal, sand, polish, mark dimples	one master per size	Check mold release before committing plaster
GEM-SC-F4-MOLD	Alto F plaster mold	1	mold box about 19.75 x 6.64 x 6.64 in	two plaster halves plus pour opening	pottery plaster	cast mold halves around sealed master	reuse for production run after seasoning	Add registration keys and split-line witness marks
GEM-SC-C5-MASTER	Soprano C scaled master	1	12.51 x 3.76 x 3.76 in envelope	10.51 in centerline, 1.76 in wide OD	sealed print/tooling board	print/CNC, seal, sand, polish, mark dimples	one master per size	Check mold release before committing plaster
GEM-SC-C5-MOLD	Soprano C plaster mold	1	mold box about 14.51 x 5.76 x 5.76 in	two plaster halves plus pour opening	pottery plaster	cast mold halves around sealed master	reuse for production run after seasoning	Add registration keys and split-line witness marks
GEM-SC-F5-MASTER	Sopranino F scaled master	1	9.87 x 3.32 x 3.32 in envelope	7.87 in centerline, 1.32 in wide OD	sealed print/tooling board	print/CNC, seal, sand, polish, mark dimples	one master per size	Check mold release before committing plaster
GEM-SC-F5-MOLD	Sopranino F plaster mold	1	mold box about 11.87 x 5.32 x 5.32 in	two plaster halves plus pour opening	pottery plaster	cast mold halves around sealed master	reuse for production run after seasoning	Add registration keys and split-line witness marks
GEM-WOOD-C5-HALVES	Soprano C CNC split-wood body halves	1 pair	12.50 x 2.50 x 1.00 in each half	10.50 in routed body envelope, 0.150 in min wall	maple/cherry/pear/boxwood	joint, plane, CNC route mirrored cavity, drill dowel registration	offcuts become fipple/block test coupons	Modern CNC router proof of concept
GEM-WOOD-G4-HALVES	Historical G4 CNC split-wood body halves	1 pair	15.50 x 3.00 x 1.10 in each half	14.00 in routed body envelope, 0.160 in min wall	maple/cherry/pear/boxwood	joint, plane, CNC route mirrored cavity, drill dowel registration	offcuts become windway and finish coupons	Good comparison against natural-horn G4
GEM-WOOD-DOWELS	Split-body registration dowels	2 to 4	0.25 dia x 1.00 in each	0.25 in dowel or removable metal pin	hardwood or steel	drill from CNC datum before cavity routing	trim flush or remove before exterior shaping	Keep outside tone-critical cavity
GEM-FINISH-COUPONS	Material/finish coupons	1 set	2.00 x 2.00 x 0.25 in samples	finished per material test	wood/ceramic/printed material	seal, finish, odor test, water test, edge-retention check	document in validation.csv	Required before alternate final-body materials

drawing-brief.md

Manufacturing drawing and technical product sketch brief.

Drawing Brief

Generated: 2026-05-02

Required Drawing Set

1. `drawings/gemshorn-family-scale.svg` - side-by-side family scale and title block.

2. `drawings/gemshorn-body-section.svg` - longitudinal body section with closed tip, wall, cavity, and fipple seat.

3. `drawings/gemshorn-voicing-detail.svg` - windway, block, labium, window, and tuning ring/vent.

4. `drawings/gemshorn-mold-schematic.svg` - two-part plaster mold, split line, keys, pour mouth, and release direction.

5. `drawings/hole-layout-template.svg` - soprano C hole positions and diameters for a drilling template.

6. `drawings/visual-bom-plate.svg` - visual BOM plate layout.

7. `drawings/gemshorn-cnc-wood-body.svg` - CNC split-wood body halves, datums, registration pins, operation notes, and tolerances.

Datums

- Datum A: labium edge / window lower edge.
- Datum B: centerline curve from labium to sealed tip.
- Datum C: dorsal plane through thumb hole and mold split witness marks.
- Datum D: wide-end block seating plane.

Critical Dimensions

- Fired and master centerline length.
- Wide-end OD/ID and block seat diameter.
- Tip OD and closed-tip minimum wall.
- Wall thickness target and acceptable range.
- Window width, window height, labium setback, windway height.
- Tone-hole center positions measured from Datum A along the centerline.
- Tone-hole final fired diameters and greenware drill-start diameters.
- Tuning ring or vent position and diameter.
- CNC split-wood body half stock, dowel-pin locations, cavity wall minimum, and glue seam datum.

Tolerances

- Tone-hole final diameter: +/-0.005 in before tuning, final by pitch.
- Window width/height: +/-0.005 in on soprano and smaller, +/-0.010 in on alto and larger.
- Windway height: +/-0.003 in for soprano/sopranino, +/-0.005 in for larger sizes.
- Wall thickness: +/-15% unless structural cracks appear.
- Noncritical exterior profile: +/-0.060 in.

Current Family Reference

ID	Key	Fired length in	Wide OD in	Wall in
GEM-SC-F3	F3	27.719	4.645	0.240
GEM-SC-C4	C4	18.500	3.100	0.204
GEM-SC-F4	F4	13.859	2.322	0.169
GEM-SC-C5	C5	9.250	1.550	0.130
GEM-SC-F5	F5	6.930	1.161	0.110

Notes

The SVGs are shop-reference drawings and review plates. Before making production molds, convert the dimensions into SolidWorks, OpenSCAD/STL, or another CAD system and verify mold release.

For the CNC wood path, verify `cad/gemshorn_split_wood_body.scad` in CAD/CAM before cutting. The SVG is a datum and operation drawing, not a toolpath source.

assembly-manual.md

Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.

Assembly Manual

Generated: 2026-05-02

Safety

- Wear a respirator for clay dust, plaster dust, sanding, and fired ceramic grinding.
- Use wet cleanup for clay and plaster; do not sweep dry dust.
- Keep plaster out of reclaim clay.
- Use eye protection when drilling bisque or fired ceramic.
- Use lead-free, food-safe-compatible glazes if the instrument may contact the mouth.

Tools

- Casting slip, plaster mold, mold straps, buckets, sieve, scale, timer.
- Calipers, flexible ruler, pin gauges, drill bits, tapered reamers, needle files.
- Chromatic tuner or microphone/spectrum app, thermometer, hygrometer.
- Hardwood for blocks, cork/leather/wax for seals, fine saw, knife, small chisels.
- CNC router, dowel-pin fixture, ball end bits, clamps, and leak-test supplies for the split-wood method.

Build Method Selection

See `build-methods.md` before starting.

- Use natural horn when authenticity is the goal.
- Use slip-cast ceramic when repeatability and consort scaling are the goal.
- Use CNC split wood when fast iteration and repairability are the goal.
- Use alternate materials only after coupon tests in `material-options.md` pass.

Phase 1: Master And Mold

1. Pick a size from `family-spec.csv`.

2. Confirm clay shrinkage with test bars.

3. Update `gemshorn-design-table.xlsx` if shrinkage differs from the default.

4. Make and seal the master.

5. Build a two-part plaster mold using `mold-and-slip-casting-plan.md`.

Phase 2: Cast Body

1. Pour slip and record drain time.

2. Release at leather hard.

3. Clean seams, mark centerline, and confirm wall thickness at the pour opening.

4. Drill undersize holes from `hole-schedule-modern.csv` or `hole-schedule-historical.csv`.

5. Trim the fipple seat and window area conservatively.

6. Slow dry to bone dry.

Phase 3: Bisque And Measure

1. Bisque fire.

2. Measure length, wide OD, hole diameters, mass, and cavity volume.

3. Record all measurements in `validation.csv`.

4. Fit a temporary block and test whether the instrument speaks.

Phase 4: Voice And Tune

1. Make the windway even and low.

2. Sharpen and stabilize the labium.

3. Tune closed note first.

4. Open holes gradually from low to high.

5. Use wax to backtrack if a hole goes sharp.

6. Record every tuning pass, temperature, and action.

Phase 5: Finish

1. Remove block before glaze unless the block is sacrificial.

2. Mask windway, labium, hole interiors, and tuning seats.

3. Glaze or burnish exterior.

4. Glaze fire.

5. Refit block, final tune, and seal mouth-contact surfaces.

CNC Split-Wood Body Method

1. Choose the C5 or G4 prototype from `build-methods.md`.

2. Prepare two straight-grained hardwood halves from `cut-list.csv`.

3. Face the mating surfaces and drill registration-pin holes.

4. Route mirrored cavity halves using `cnc/wood-body-cnc-plan.md`.

5. Engrave or mark fipple, tone-hole, and datum references.

6. Dry-fit the body with dowels and perform the leak test in `validation.csv`.

7. Glue the body only after the cavity volume and leak test pass.

8. Shape the exterior, then cut the fipple/window by hand.

9. Drill holes undersize and tune low to high.

10. Record final volume, pitch, cents error, and finish notes in `validation.csv`.

Maintenance

- Keep removable blocks dry and replaceable.
- Avoid soaking fired ceramic bodies unless the clay/glaze combination is proven.
- Store with a cork or cloth spacer so the labium is protected.
- Keep a small wax kit for tuning during outdoor or temperature-variable use.

validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

instrument_id	check	target	measured	tolerance	environment	result	action	measured_hz	cents_error	tuner	measurement_date
GEM-SC-F3	closed tonic frequency	174.61 Hz		+/-15 cents prototype	record temp F, humidity		adjust cavity/window/vent before hole tuning				
GEM-SC-F3	fired centerline length	27.719 in		+/-0.060 in	after glaze if glazed		update shrinkage factor if systematic				
GEM-SC-F3	fired internal volume	2402 ml		+/-5%	water-fill, dry after		retune window/vent model				
GEM-SC-C4	closed tonic frequency	261.63 Hz		+/-15 cents prototype	record temp F, humidity		adjust cavity/window/vent before hole tuning				
GEM-SC-C4	fired centerline length	18.500 in		+/-0.060 in	after glaze if glazed		update shrinkage factor if systematic				
GEM-SC-C4	fired internal volume	650 ml		+/-5%	water-fill, dry after		retune window/vent model				
GEM-SC-F4	closed tonic frequency	349.23 Hz		+/-15 cents prototype	record temp F, humidity		adjust cavity/window/vent before hole tuning				
GEM-SC-F4	fired centerline length	13.859 in		+/-0.060 in	after glaze if glazed		update shrinkage factor if systematic				
GEM-SC-F4	fired internal volume	260 ml		+/-5%	water-fill, dry after		retune window/vent model				
GEM-SC-C5	closed tonic frequency	523.25 Hz		+/-15 cents prototype	record temp F, humidity		adjust cavity/window/vent before hole tuning				
GEM-SC-C5	fired centerline length	9.250 in		+/-0.060 in	after glaze if glazed		update shrinkage factor if systematic				
GEM-SC-C5	fired internal volume	72 ml		+/-5%	water-fill, dry after		retune window/vent model				
GEM-SC-F5	closed tonic frequency	698.46 Hz		+/-15 cents prototype	record temp F, humidity		adjust cavity/window/vent before hole tuning				
GEM-SC-F5	fired centerline length	6.930 in		+/-0.060 in	after glaze if glazed		update shrinkage factor if systematic				
GEM-SC-F5	fired internal volume	28 ml		+/-5%	water-fill, dry after		retune window/vent model				
GEM-SC-F3	H1 final diameter	0.3409 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F3	H2 final diameter	0.3936 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F3	H3 final diameter	0.2857 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F3	H4 final diameter	0.4914 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F3	H5 final diameter	0.5723 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F3	H6 final diameter	0.6692 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F3	H7 final diameter	0.4727 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				

GEM-SC-F3	T1 final diameter	0.8530 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C4	H1 final diameter	0.2347 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C4	H2 final diameter	0.2699 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C4	H3 final diameter	0.1977 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C4	H4 final diameter	0.3345 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C4	H5 final diameter	0.3875 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C4	H6 final diameter	0.4504 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C4	H7 final diameter	0.3222 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C4	T1 final diameter	0.5687 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F4	H1 final diameter	0.1780 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F4	H2 final diameter	0.2043 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F4	H3 final diameter	0.1502 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F4	H4 final diameter	0.2525 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F4	H5 final diameter	0.2919 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F4	H6 final diameter	0.3386 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F4	H7 final diameter	0.2434 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F4	T1 final diameter	0.4258 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C5	H1 final diameter	0.1208 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C5	H2 final diameter	0.1383 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C5	H3 final diameter	0.1021 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C5	H4 final diameter	0.1703 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C5	H5 final diameter	0.1963 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C5	H6 final diameter	0.2270 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				

GEM-SC-C5	H7 final diameter	0.1643 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-C5	T1 final diameter	0.2840 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F5	H1 final diameter	0.0917 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F5	H2 final diameter	0.1048 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F5	H3 final diameter	0.0777 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F5	H4 final diameter	0.1287 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F5	H5 final diameter	0.1480 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F5	H6 final diameter	0.1707 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F5	H7 final diameter	0.1242 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-SC-F5	T1 final diameter	0.2127 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-HIST-G4	F1 final diameter	0.1594 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-HIST-G4	F2 final diameter	0.1828 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-HIST-G4	F3 final diameter	0.1345 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-HIST-G4	F4 final diameter	0.2257 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-HIST-G4	T1 final diameter	0.5445 in		by pitch, do not oversize	after bisque and final tune		open gradually or wax if sharp				
GEM-WOOD-C5	split-body cavity leak test	no bubbles at 2 minute low-pressure test		pass/fail	shop bench before exterior shaping		seal seam or remake glue-up				
GEM-WOOD-C5	measured internal volume	match C5 target within +/-5%		+/-5%	water-fill after sealed glue-up		update Helmholtz model and hole schedule				
GEM-WOOD-G4	split-body cavity leak test	no bubbles at 2 minute low-pressure test		pass/fail	shop bench before exterior shaping		seal seam or remake glue-up				
GEM-MATERIAL-COUPON	finish odor and mouth-contact screen	no tacky odor or surface shedding after cure		pass/fail	after full cure at shop temperature		reject finish or move finish away from mouth-contact areas				
GEM-MATERIAL-COUPON	labium edge retention	coupon edge stays crisp after finish/wet-dry handling		visual and caliper check	after finish and wipe-down		change material or mask voicing edges				

supplier-rfq.md

Supplier email/request-for-quote starter.

Supplier RFQ Draft

Subject: RFQ - slip-cast ceramic gemshorn prototypes and plaster mold materials

Hello,

I am prototyping a family of slip-cast ceramic gemshorns, a small fipple/vessel-flute instrument. I am looking for pricing, lead time, and technical guidance for the following:

1. White stoneware or porcelain casting slip suitable for thin musical-instrument bodies.

2. Pottery plaster or equivalent mold plaster for two-part molds.

3. Optional 3D print or CNC master service for sealed masters up to about 31.5 in long and 5.3 in wide.

4. Small hardwood, cork, leather, or silicone sheet suitable for removable fipple blocks and seals.

5. Brass or stainless thin-wall tubing/strip for optional rotating tuning collars.

Please quote unit price, volume price, minimum order, lead time, shipping estimate, data sheets, shrinkage range, recommended firing cone, and any substitutions you recommend.

The current prototype set includes: F3, C4, F4, C5, F5. Critical requirements are stable shrinkage, clean release from plaster molds, and a fired body that can hold crisp tone-hole and voicing edges.

Thank you,

Tony

visual-bom-brief.md

Art direction for an image-forward visual BOM.

Visual BOM Brief

Goal

Create an image-forward resource plate for the slip-cast gemshorn family: finished instrument, mold, master, clay body, fipple block, tuning ring, tools, and measurement/tuning kit.

Layout

- Header: "Slip-Cast Gemshorn Family", quote/build date, revision, estimated cost status.
- Hero: family scale view with bass F through sopranino F.
- Exploded center: ceramic body, removable block, leather/cork seal, tuning ring/collar, hole/tuning kit.
- Table: item, qty, make/buy, material/spec, source/search term, drawing ref, cost status.
- Footer: shrinkage assumption, clay body TBD, source/history note.

Image Status

- `drawings/visual-bom-plate.svg` is a generated vector placeholder.
- Replace placeholders with real photos after the first cast, master, mold, and fipple block exist.
- Do not use generated images for critical dimensions.

wolfram-starter.wl

Wolfram starter for physics, optimization, visualization, and validation.

```
(* Slip-Cast Gemshorn Family Wolfram Starter *)
(* Generated 2026-05-02. Paste into Wolfram Desktop or Cloud. *)

ClearAll["Global`*"];

speedOfSoundInPerS = 13549.0;
shrinkage = 0.12;

family = {
  <|"ID" -> "GEM-SC-F3", "Key" -> "F3", "MIDI" -> 53, "LengthIn" -> 27.7187, "VolumeIn3" -> 146.5677, "WindowAreaIn2" ->
0.54058, "WallIn" -> 0.2400|>,
  <|"ID" -> "GEM-SC-C4", "Key" -> "C4", "MIDI" -> 60, "LengthIn" -> 18.5000, "VolumeIn3" -> 39.6387, "WindowAreaIn2" ->
0.24080, "WallIn" -> 0.2040|>,
  <|"ID" -> "GEM-SC-F4", "Key" -> "F4", "MIDI" -> 65, "LengthIn" -> 13.8593, "VolumeIn3" -> 15.8811, "WindowAreaIn2" ->
0.13514, "WallIn" -> 0.1691|>,
  <|"ID" -> "GEM-SC-C5", "Key" -> "C5", "MIDI" -> 72, "LengthIn" -> 9.2500, "VolumeIn3" -> 4.3665, "WindowAreaIn2" -> 0.06020,
"WallIn" -> 0.1300|>,
  <|"ID" -> "GEM-SC-F5", "Key" -> "F5", "MIDI" -> 77, "LengthIn" -> 6.9297, "VolumeIn3" -> 1.6964, "WindowAreaIn2" -> 0.03379,
"WallIn" -> 0.1100|>
};

midiToFreq[m_] := 440*2^((m - 69)/12);
centsError[measured_, target_] := 1200*Log[2, measured/target];
holeLeff[diameter_, wall_] := wall + 0.85*(diameter/2);
conductance[diameter_, wall_] := Pi*(diameter/2)^2/holeLeff[diameter, wall];
helmholtzHz[conductanceTotal_, volumeIn3_] :=
  speedOfSoundInPerS/(2*Pi)*Sqrt[conductanceTotal/volumeIn3];

Dataset[family]

Manipulate[
 Module[{freq = midiToFreq[midi], scale, volume, windowArea, wall, windowLeff, gBase, predicted},
  scale = midiToFreq[72]/freq;
  wall = Min[0.24, Max[0.11, 0.13*scale^0.65]];
  windowArea = (0.43*scale)*(0.14*scale);
  windowLeff = wall + 0.85*Sqrt[windowArea/Pi];
  gBase = windowArea/windowLeff;
  volume = 4.366467*scale^3;
  predicted = helmholtzHz[gBase, volume];
  Column[{
    Row[{"Target Hz: ", NumberForm[freq, {6, 2}]}],
    Row[{"Predicted closed Hz: ", NumberForm[predicted, {6, 2}]}],
    Row[{"Cents error: ", NumberForm[centsError[predicted, freq], {6, 1}]}],
    Row[{"Scale factor: ", NumberForm[scale, {5, 3}]}],
    Row[{"Volume in3: ", NumberForm[volume, {7, 3}]}],
    Row[{"Window area in2: ", NumberForm[windowArea, {6, 4}]}]
  }]
 ],
 {{midi, 72, "Closed-note MIDI"}, 53, 77, 1}
]

(* Import validation.csv after prototypes exist:
validation = Import["validation.csv", "Dataset"];
*)
```

README.md

Project artifact.

Gemshorn

`![Gemshorn family scale drawing](drawings/gemshorn-family-scale.svg)`

Generated/refreshed: 2026-05-06

This repo is the build documentation for an authentic/historically informed gemshorn and a modern slip-cast gemshorn consort. A gemshorn is a closed horn-shaped vessel flute: the pointed end is sealed, the wide end carries the fipple/block and voicing window, and pitch is governed primarily by chamber volume, window/vent conductance, and opened tone-hole area.

The repo deliberately separates the strict historical path from modern production paths. Natural horn is the authenticity reference; slip-cast ceramic is the repeatable family path; CNC-routed split wood is a modern prototype path for quick iteration and teaching.

Quick Start

1. Read `design.md`, `build-methods.md`, and `authenticity-notes.md`.

2. Open `gemshorn-design-table.xlsx` and replace shrinkage with your clay test-bar value.

3. Build the CNC-routed wood `GEM-SC-C5` if you want the fastest fipple/tuning test.

4. Build the slip-cast ceramic `GEM-SC-C5` to validate shrinkage and mold workflow.

5. Build the natural-horn `GEM-HIST-G4` to establish the authentic reference voice.

6. Update `validation.csv` after each prototype before scaling the full family.

Build Methods

Method	Purpose	Main Files
Natural horn	Most historically conservative build path	`authentic-horn-build-plan.md`, `horn-blank-spec.csv`, `hole-schedule-historical.csv`
Slip-cast ceramic	Repeatable production family	`mold-and-slip-casting-plan.md`, `family-spec.csv`, `hole-schedule-modern.csv`
CNC-routed split wood	Modern fast prototype / teaching body	`cnc/wood-body-cnc-plan.md`, `cad/gemshorn_split_wood_body.scad`, `drawings/gemshorn-cnc-wood-body.svg`
Alternate materials	Research and prototyping candidates	`material-options.md`, `risks.md`, `validation.csv`

File Map

File	Purpose
design.md	Project intent, acoustic model, assumptions, and file map.
build-methods.md	Real horn, slip-cast ceramic, CNC wood, and research-material build paths.
material-options.md	Candidate body/block/finish materials and safety gates.
authenticity-notes.md	Historical/provenance notes and source links.
authentic-horn-build-plan.md	Natural-horn authentic build workflow.
horn-blank-spec.csv	Horn blank buying/selection requirements.
gemshorn-design-table.xlsx	Parametric spreadsheet with formulas.
family-spec.csv	Size family dimensions.
hole-schedule-modern.csv	Modern consort hole schedule.
hole-schedule-historical.csv	Historically informed pilot hole schedule.
mold-and-slip-casting-plan.md	Master, mold, and casting workflow.
assembly-manual.md	Shop sequence from casting to final tuning.
risks.md	Red-team risk register with tests and mitigations.
photo-shotlist.md	Required photos for the v4 build-log site and future deck refreshes.
drawings/	SVG drawings, visual BOM plate, and CNC wood-body drawing.
cad/	OpenSCAD master-shape starters for ceramic and split-wood paths.

cnc/	CNC operation plan, setup sheet, operations table, and wood-body CNC notes.
images/	Placeholder folder and naming rules for future real build photos.
data/	Material-test matrix and future measured-data tables.
wolfram-starter.wl	Lightweight physics starter.
wolfram/instrument-model.wl	v4.2 Wolfram model package.
site/index.html	Static build-log site.
print-packet.html / print-packet.pdf	Shop packet for browser or print use.
capstone-deck.md / capstone-deck.pptx	Capstone orientation deck.

First Build Recommendation

Start with a wood or ceramic `GEM-SC-C5` because it is small enough to make quickly and large enough to work by hand. After that, make `GEM-HIST-G4` in real horn to test the historically informed four-hole path. Expand to bass/tenor only after measured shrinkage, volume, leak, and tuning corrections are in `validation.csv`.

v4 Deliverable Status

- Parametric design: `gemshorn-design-table.xlsx`
- Build methods: `build-methods.md`
- BOM/sourcing/cut/validation: `bom.csv`, `sourcing.csv`, `cut-list.csv`, `validation.csv`
- Drawings: `drawings/*.svg`
- CNC plan: `cnc/cnc-plan.json`, `cnc/operations.csv`, `cnc/setup-sheet.md`
- Wolfram package: `wolfram/instrument-model.wl`
- Risks: `risks.md`
- Photo pipeline: `photo-shotlist.md`
- Build-log site: `site/index.html`
- Capstone and print packet: `capstone-deck.*`, `print-packet.*`

authentic-horn-build-plan.md

Project artifact.

Authentic Natural-Horn Gemshorn Build Plan

Generated: 2026-05-02

Intent

This is the most historically conservative build path in the packet: a short curved horn body with a fipple/block at the wide end, closed pointed tip, limited hole count, and soft vessel-flute voice. Use domesticated cow, ox, goat, or sheep horn. Do not source endangered chamois or wild-animal horn.

Target Prototype

- Prototype ID: `GEM-HIST-G4`
- Suggested starting pitch: `G4` at `392.00 Hz`
- Design centerline length: `12.35 in`
- Wide OD target: `2.07 in`
- Internal volume target: `180 ml`
- Hole system: four front holes plus one back/thumb vent, with the third front hole optional for a stricter three-front-plus-thumb reconstruction.

Horn Blank Selection

- Pick a horn with a naturally sealed or solid tip and enough length to trim back to pitch.
- Avoid cracks, deep delamination, insect damage, and extremely thin walls near the wide end.
- Prefer a smooth curve that lets the front holes sit under the hands without wrist strain.
- Wide-end OD should be near the target, but volume matters more than exterior size.
- Buy at least three blanks for the first working instrument; horn geometry varies too much for one-shot confidence.

Cleaning And Prep

1. Remove loose core material mechanically.

2. Degrease with warm water and mild detergent. Avoid boiling unless you already know the horn tolerates it.

3. Dry completely.

4. Scrape, sand, or ream the interior only enough to remove soft material and stabilize the cavity.

5. Keep the tip closed. If the tip opens accidentally, plug it permanently and document the added volume.

6. Measure water-fill volume and update the design table notes.

Wide-End Fipple

1. Square and flatten the wide-end seating plane enough to accept a block.

2. Fit a removable hardwood or plaster block.

3. Cut a low, even windway.

4. Cut the voicing window in the horn wall and form a crisp labium edge.

5. Use leather, cork, wax, or thread wrap to seal around the block.

6. Add a small tuning vent/ring only after the closed note speaks.

Historical Hole Pilot

Hole	Face	From labium in	Final dia in	Green/pilot dia in	Target
F1	front distal	5.309	0.159	0.181	+2 semitones

F2	front	4.198	0.183	0.208	+4 semitones
F3	front	3.334	0.135	0.153	+5 semitones
F4	front proximal	2.531	0.226	0.256	+7 semitones
T1	back thumb	2.531	0.544	0.619	+12 semitones

For animal horn, use these as layout starting points, then tune empirically:

- Drill undersize first.
- Tune low to high.
- Lower over-sharp notes with wax while you learn the horn's response.
- Record the final hole diameter, position, and pitch in `validation.csv`.

Tuning Workflow

1. Make the closed note speak with no finger holes open.

2. Adjust the fipple/block and window before drilling tone holes.

3. Drill F1 undersize, tune it, then move upward.

4. If the octave/back vent is too aggressive, reduce it with wax or make it a rotating collar vent.

5. Final-polish the mouth area only after tuning is stable.

Finish

- Polish horn exterior by sanding through fine grits and buffing.
- Oil lightly only after all adhesive and block materials are compatible.
- Keep the windway dry, crisp, and free from oily residue.
- Label the finished instrument as a historically informed reconstruction, not a museum copy.

authenticity-notes.md

Project artifact.

Authenticity Notes

Generated: 2026-05-02

What Counts As Authentic Here

For this packet, "authentic gemshorn" means a historically informed instrument rather than a museum-certified copy. The available evidence supports these constraints:

- The body is a short curved horn form, traditionally animal horn and plausibly ceramic in at least one late medieval find.
- It is blown from the wide/open end, which is closed by a fipple/block and voicing window.
- The pointed end stays closed, so the instrument behaves as a stopped inverted cone or vessel flute.
- The range is limited because it does not overblow like a recorder.
- The safest historical hole count is small. Four-hole evidence is stronger than the modern seven-front-hole consort layout.

What Is Modern In This Packet

The slip-cast consort family uses modern reconstruction logic:

- Repeatable molded ceramic bodies rather than individual animal horns.
- Seven front holes plus a back thumb/tuning vent for ensemble usability.
- Equal-temperament C/F family targets.
- Spreadsheet-derived hole diameters and scaling.

That modern line is useful and buildable, but it should not be labeled as a direct 15th-century copy.

Historically Informed Build Choices

- Make the first historical pilot in G4 because modern measured reconstructions around that register provide a dimensional sanity check.
- Use a waxed or replaceable tuning ring/vent near the voicing area. Historical and modern sources describe tuning by partially covering a vent/ring or thumb hole.
- Keep a soft, low-wind sound. Overblown recorder behavior is not expected.
- Keep exterior decoration restrained: polished horn, burnished ceramic, leather wrap, or simple incised marks. Avoid shiny glaze near the voicing.

Source Notes

- [Horace Fitzpatrick, The Gemshorn: a Reconstruction](https://www.cambridge.org/core/journals/proceedings-of-the-royal-musical-association/article/gemshorn-a-reconstruction/0AD60F0FE7748111C8455FA2F4DD1CC3): Defines the gemshorn as a fipple instrument made from animal horn and calls out the stopped inverted cone bore. Used as the boundary-condition anchor.
- [Early Music Muse, The gemshorn: a short history](https://earlymusicmuse.com/gemshorn/): Summarizes the open-end fipple construction, four-hole historical evidence, and the 1455 clay gemshorn-like find. Used for authenticity notes.
- [Cincinnati Early Music, Medieval Gemshorn](https://cincinnatiearlymusic.com/gemshorn.html): Contrasts historical four-hole evidence with modern recorder-like reconstructions and consort families. Used to separate historical and modern lines.
- [Baltimore Recorders, Information about the Gemshorn](https://www.baltimorerecorders.org/gemshorns.html): Describes front holes, thumb hole, whistle/fipple, tuning ring, and the lack of an outlet other than whistle and fingerholes. Used for construction details.
- [Institut fuer Musikforschung Wuerzburg, Lo 29-31 Gemshoerner](https://www.musikwissenschaft.uni-wuerzburg.de/musikinstrumente/bestand/inventarliste/lo29-31-gemshoerner/): Gives measured modern HFK gemshorns in g, d1, and g1 and cautions that modern gemshorns are reconstruction products. Used as dimensional sanity check.

Provenance Warning

No file in this packet claims to reproduce an extant original gemshorn exactly. The historical record is sparse; this packet gives a traceable and tunable workshop path with explicit assumptions.

build-methods.md

Project artifact.

Gemshorn Build Methods

Generated: 2026-05-06

Build Method Matrix

Method	Authenticity	Repeatability	Tooling	Best First Size	Main Risk	Recommended Use
Natural horn	Highest historical fit	Low, each blank differs	Hand tools, drill press, block fitting tools	`GEM-HIST-G4`	Horn volume/curve varies and forces empirical tuning	Make the historically informed reference instrument
Slip-cast ceramic	Historically plausible clay lineage, modern process	High after shrinkage data	Master, plaster mold, casting slip, kiln	`GEM-SC-C5`	Shrinkage and voicing edge movement	Make the repeatable consort family
CNC-routed wood split body	Modern adaptation	Medium to high	CNC router, flip jig, clamps, glue-up, fipple tools	`GEM-SC-C5` or `GEM-HIST-G4`	Glue seam leaks and routed cavity volume error	Make fast prototypes and serviceable teaching instruments
3D printed body	Modern prototype only	High	SLA/FDM printer, sealant, post-processing	`GEM-SC-C5`	Mouth safety, porosity, rough windway	Fit/ergonomics tests before committing molds
Carved hardwood solid horn	Modern craft adaptation	Low to medium	Carving, gouges, drills, fipple tools	`GEM-HIST-G4`	Internal volume hard to control	One-off art instrument
Metal spun/formed shell	Modern experimental	Medium after tooling	Metal forming or fabrication	Alto/tenor only	Condensation, sharp edges, tuning vent noise	Research or sculpture, not first playable build
Gourd/calabash or seed pod	Folk-adjacent experiment	Low	Hand tools, sealing, block fitting	Small high register	Cracking and inconsistent volume	Acoustic experiment, not production
Resin/composite cast	Modern prototype	High	Silicone mold or printed mold, resin safety controls	`GEM-SC-C5`	Mouth-contact safety and off-gassing	Non-mouth-contact display or sealed prototype only

Method 1: Natural-Horn Gemshorn

Use `authentic-horn-build-plan.md` and `horn-blank-spec.csv`.

1. Select several domesticated cow, ox, goat, or sheep horn blanks.

2. Clean and stabilize the cavity while keeping the tip closed.

3. Fit a removable hardwood or plaster block at the wide end.

4. Cut the windway and labium before drilling tone holes.

5. Tune the closed note by block/window/vent behavior.

6. Drill holes undersize, tune low to high, and record final values in `validation.csv`.

This method should be labeled "historically informed reconstruction." Do not claim a museum copy unless the exact source instrument and dimensions are known.

Method 2: Slip-Cast Ceramic Family

Use `mold-and-slip-casting-plan.md`, `family-spec.csv`, and `hole-schedule-modern.csv`.

1. Confirm clay shrinkage with test bars.

2. Make the `GEM-SC-C5` master first.

3. Cast a two-part plaster mold around a sealed master.

4. Cast one body, bisque, measure volume, and tune.

5. Update the design table before scaling to larger family members.

6. Keep the windway, labium, tone-hole interiors, and block seat unglazed.

This is the best production path once shrinkage and tuning corrections are measured.

Method 3: CNC-Routed Split-Wood Body

Use `cad/gemshorn_split_wood_body.scad`, `drawings/gemshorn-cnc-wood-body.svg`, and `cnc/wood-body-cnc-plan.md`.

1. Mill two matching wood halves oversized.

2. Route registration-pin holes.

3. Route mirrored internal cavity halves with a ball end or round-nose bit.

4. Add tone-hole pilot marks and fipple/window layout.

5. Dry-fit with dowel pins and test the closed cavity for leaks.

6. Glue with a thin, continuous, reversible-enough seam strategy for prototypes.

7. Shape exterior, cut the voicing, and tune as a vessel flute.

This path is not historically authentic, but it is valuable because the body can be measured, repaired, and iterated without kiln or horn-blank variability.

Method 4: Prototype And Research Materials

Use `material-options.md` before committing to any material outside horn, ceramic, or wood. The material must pass three gates:

- It must be safe around the mouth after finish.
- It must hold a crisp labium edge and stable tone-hole edges.
- It must maintain airtight seams and predictable cavity volume.

Recommended Build Sequence

1. CNC-routed wood `GEM-SC-C5` to prove the fipple and Helmholtz schedule quickly.

2. Slip-cast ceramic `GEM-SC-C5` to validate shrinkage and mold process.

3. Natural-horn `GEM-HIST-G4` to establish the authentic reference voice.

4. Expand the ceramic family from C5 outward after measured corrections exist.

family-spec.csv

Project artifact.

id	name	key	midi	freq_hz	scale	line	fired_centerline_length_in	master_centerline_length_in	fired_wide_od_in	master_wide_od_in	fired_tip_od_in	fire
GEM-SC-F3	Bass F	F3	53	174.6141	2.9966	modern slip-cast consort	27.7187	31.4985	4.6448	5.2781	1.0788	0.24
GEM-SC-C4	Tenor C	C4	60	261.6256	2.0	modern slip-cast consort	18.5	21.0227	3.1	3.5227	0.72	0.20
GEM-SC-F4	Alto F	F4	65	349.2282	1.4983	modern slip-cast consort	13.8593	15.7493	2.3224	2.6391	0.5394	0.16
GEM-SC-C5	Soprano C	C5	72	523.2511	1.0	modern slip-cast consort baseline	9.25	10.5114	1.55	1.7614	0.36	0.13
GEM-SC-F5	Sopranino F	F5	77	698.4565	0.7492	modern slip-cast consort	6.9297	7.8746	1.1612	1.3195	0.2697	0.11
GEM-HIST-G4	Historical Clay/Horn Archetype	G4	67	391.9954	1.3348	historically informed 4 front plus thumb study	12.3473	14.031	2.069	2.3511	0.4805	0.15

hole-schedule-historical.csv

Project artifact.

instrument_id	key	hole	face	target_interval_semitones	previous_interval_semitones	position_pct_from_labium	fired_position_from_labium_in	master_position_from_lal
GEM-HIST-G4	G4	F1	front distal	2	0	0.43	5.3093	6.0333
GEM-HIST-G4	G4	F2	front	4	2	0.34	4.1981	4.7705
GEM-HIST-G4	G4	F3	front	5	4	0.27	3.3338	3.7884
GEM-HIST-G4	G4	F4	front proximal	7	5	0.205	2.5312	2.8764
GEM-HIST-G4	G4	T1	back thumb	12	7	0.205	2.5312	2.8764

hole-schedule-modern.csv

Project artifact.

instrument_id	key	hole	face	target_interval_semitones	previous_interval_semitones	position_pct_from_labium	fired_position_from_labium_in	master_position_from_lal
GEM-SC-F3	F3	H1	front distal	2	0	0.479	13.2772	15.0878
GEM-SC-F3	F3	H2	front	4	2	0.424	11.7527	13.3554
GEM-SC-F3	F3	H3	front	5	4	0.369	10.2282	11.6229
GEM-SC-F3	F3	H4	front	7	5	0.316	8.7591	9.9535
GEM-SC-F3	F3	H5	front	9	7	0.263	7.29	8.2841
GEM-SC-F3	F3	H6	front	11	9	0.21	5.8209	6.6147
GEM-SC-F3	F3	H7	front proximal	12	11	0.157	4.3518	4.9453
GEM-SC-F3	F3	T1	back thumb	14	12	0.157	4.3518	4.9453
GEM-SC-C4	C4	H1	front distal	2	0	0.479	8.8615	10.0699
GEM-SC-C4	C4	H2	front	4	2	0.424	7.844	8.9136
GEM-SC-C4	C4	H3	front	5	4	0.369	6.8265	7.7574
GEM-SC-C4	C4	H4	front	7	5	0.316	5.846	6.6432
GEM-SC-C4	C4	H5	front	9	7	0.263	4.8655	5.529
GEM-SC-C4	C4	H6	front	11	9	0.21	3.885	4.4148
GEM-SC-C4	C4	H7	front proximal	12	11	0.157	2.9045	3.3006
GEM-SC-C4	C4	T1	back thumb	14	12	0.157	2.9045	3.3006
GEM-SC-F4	F4	H1	front distal	2	0	0.479	6.6386	7.5439
GEM-SC-F4	F4	H2	front	4	2	0.424	5.8764	6.6777
GEM-SC-F4	F4	H3	front	5	4	0.369	5.1141	5.8115

GEM-SC-F4	F4	H4	front	7	5	0.316	4.3796	4.9768
GEM-SC-F4	F4	H5	front	9	7	0.263	3.645	4.1421
GEM-SC-F4	F4	H6	front	11	9	0.21	2.9105	3.3073
GEM-SC-F4	F4	H7	front proximal	12	11	0.157	2.1759	2.4726
GEM-SC-F4	F4	T1	back thumb	14	12	0.157	2.1759	2.4726
GEM-SC-C5	C5	H1	front distal	2	0	0.479	4.4307	5.0349
GEM-SC-C5	C5	H2	front	4	2	0.424	3.922	4.4568
GEM-SC-C5	C5	H3	front	5	4	0.369	3.4133	3.8787
GEM-SC-C5	C5	H4	front	7	5	0.316	2.923	3.3216
GEM-SC-C5	C5	H5	front	9	7	0.263	2.4327	2.7645
GEM-SC-C5	C5	H6	front	11	9	0.21	1.9425	2.2074
GEM-SC-C5	C5	H7	front proximal	12	11	0.157	1.4523	1.6503
GEM-SC-C5	C5	T1	back thumb	14	12	0.157	1.4523	1.6503
GEM-SC-F5	F5	H1	front distal	2	0	0.479	3.3193	3.7719
GEM-SC-F5	F5	H2	front	4	2	0.424	2.9382	3.3388
GEM-SC-F5	F5	H3	front	5	4	0.369	2.557	2.9057
GEM-SC-F5	F5	H4	front	7	5	0.316	2.1898	2.4884
GEM-SC-F5	F5	H5	front	9	7	0.263	1.8225	2.071
GEM-SC-F5	F5	H6	front	11	9	0.21	1.4552	1.6537
GEM-SC-F5	F5	H7	front proximal	12	11	0.157	1.088	1.2363

GEM-SC-F5	F5	T1	back thumb	14	12	0.157	1.088	1.2363

horn-blank-spec.csv

Project artifact.

use_case	preferred_material	blank_centerline_length_in	wide_end_od_in	minimum_wall_in	tip_condition	reject_if	notes
Historical G4 pilot	domesticated cow, ox, goat, or sheep horn	13.35 to 16.35	1.76 to 2.59	0.070 near wide end after cleaning	closed or plug-friendly	cracked, delaminated, moldy core, thin/transparent wall at voicing, severe twist under hands	Buy several blanks; tune by measured volume and final holes
Small high historical pilot	goat or sheep horn	7.5 to 10.5	1.2 to 1.8	0.055 near wide end after cleaning	closed	too small for removable block or stable windway	Good fast test for fipple and hole-count behavior
Lower/alto horn pilot	cow or ox horn	13.5 to 18.0	1.8 to 3.0	0.080 near wide end after cleaning	closed or permanently pluggable	wide end too oval to seal with a block	Likely more playable than very small horns for first tuning tests

material-options.md

Project artifact.

Gemshorn Material Options

Generated: 2026-05-06

Material Selection Rules

A gemshorn body is a closed vessel flute. The material mostly affects durability, edge stability, moisture behavior, weight, and finish, while pitch is dominated by chamber volume, window/vent area, tone-hole conductance, and fipple behavior.

Use any alternate material only after a small coupon proves:

- The labium edge can stay crisp.
- The windway can stay smooth and airtight.
- The material is safe for mouth contact after finish.
- The body can be cleaned and dried without swelling, softening, or shedding dust.

Candidate Materials

Material	Build Path	Pros	Risks	Recommendation
Cow/ox/goat/sheep horn	Natural horn	Strong authenticity, beautiful surface, naturally curved closed cone	Variable volume, odor/dust while working, ethical sourcing	Best authentic reference path
White stoneware casting slip	Slip-cast	Durable, repeatable, historically plausible clay form	Shrinkage, warping, glaze shifts pitch	Best production family path
Porcelain casting slip	Slip-cast	Fine detail, crisp holes, clean white body	More brittle, shrinkage can be high	Good after stoneware pilot
Maple, cherry, pear, boxwood	CNC split wood	Stable, machinable, good edge detail	Glue seam leaks, humidity movement	Best CNC router path
Cedar or cypress	CNC split wood/block	Easy to work, light, traditional windway/block feel	Soft labium, dents, resin/finish concerns	Good for removable blocks, not first body
Bamboo or cane	Hand-built tube/horn-like adaptation	Light, musical, low cost	Natural tube is not horn-shaped and tip sealing is awkward	Good experiment, not authentic gemshorn
Gourd/calabash	Hand-built vessel	Folk resonance, light	Cracks, inconsistent wall, hard fipple seat	Experimental only
SLA resin	Printed prototype	Accurate shape, fast iteration	Mouth safety and post-cure uncertainty	Use for masters or non-mouth-contact prototypes
PETG/PLA	FDM prototype/master	Cheap, fast, good mold master after sealing	Layer leaks, rough windway, heat sensitivity	Use as mold master, not final instrument
Machinable wax/tooling board	CNC master	Stable for mold masters	Not final instrument material	Excellent for plaster mold masters
Aluminum/brass/copper	Metal shell	Durable, striking look	Condensation, thermal feel, sharp-edge risk, fabrication complexity	Advanced research only
Acrylic/polycarbonate	CNC/laser/formed prototype	Transparent tuning demo	Mouth safety, solvent crazing, brittle chips	Teaching display only unless safety proven
Bone/antler	Hand-built	Dense and visually related to horn	Ethical sourcing, dust hazards, small sizes	Avoid for first builds

Preferred Material Stack

Component	Preferred	Acceptable Substitute	Avoid
Authentic body	Domesticated horn	Burnished ceramic historical pilot	Wild/endangered horn
Ceramic production body	White stoneware slip	Porcelain slip	Unknown shrinkage hobby clay
CNC wood body	Maple, cherry, pear, boxwood	Walnut, birch, beech	Open-pore oily woods near mouth
Fipple/block	Pear, maple, cedar	Plaster or sealed hardwood	Soft crumbly wood, toxic finishes
Gasket/seal	Cork, leather, beeswax	Food-safe silicone	Permanent brittle epoxy at tuning joints
Finish near mouth	Burnish, shellac after safety review, food-safe oil/wax	Exterior-only glaze away from windway	Heavy glaze in windway or hole interiors

Material Experiments Worth Doing

1. Ceramic body with wood removable block.
2. CNC maple split body with cork gasket and reversible block.
3. Transparent acrylic section model for teaching airflow and cavity behavior.
4. 3D printed master versus CNC tooling-board master shrinkage comparison.
5. Porcelain versus stoneware tuning shift after glaze fire.

mold-and-slip-casting-plan.md

Project artifact.

Mold And Slip-Casting Plan

Generated: 2026-05-02

Build Strategy

Use a sealed master for each size and make a two-part plaster mold with the split line along the outside curvature. The wide end acts as the pour opening and later receives the fipple/block system.

Master Prep

1. Print or CNC the master from `cad/gemshorn_family.scad` dimensions.

2. Scale all fired dimensions by `1/(1 - shrinkage)`. The default packet value is `1.1364`.

3. Add cast-in centerline marks, hole dimples, window outline, split-line marks, and registration witness marks.

4. Seal print layers with epoxy/primer/shellac until plaster cannot grip the surface.

5. Polish the master enough that a leather-hard cast releases without tearing.

Mold Layout

- Two main plaster halves.
- Split line follows the low-stress side of the curved horn, avoiding the labium edge.
- Add at least three registration keys per side.
- Leave a broad pour mouth at the wide end.
- Add a removable insert or post only if your chosen shape traps the tip.
- Provide a small witness dimple at each hole location. Do not rely on cast-in final hole sizes for first prototypes.

Casting Cycle

1. Dry mold fully before first pour.

2. Pour well-mixed casting slip through the wide end.

3. Time wall growth with test coupons, then drain.

4. Release at soft leather hard.

5. Open/clean the pour end and trim the fipple seat.

6. Drill pilot holes undersize using the greenware diameters in `hole-schedule-*.csv`.

7. Dry slowly with the wide end loosely covered for the first day.

8. Bisque fire.

9. Measure shrinkage, volume, mass, hole diameters, and closed-note pitch.

10. Tune by carefully opening holes. Keep all tuning notes in `validation.csv`.

Fipple And Voicing Options

- **Authentic horn-style:** wood or plaster block fitted into the wide end; leather wrap forms or seals the windway.
- **Ceramic production:** slip-cast body plus fitted hardwood block, removable for service and tuning.
- **All ceramic:** possible, but higher risk. It requires a precise windway insert and more shrinkage control.

Clay And Finish Rules

- Do not glaze inside the windway or on the labium edge.
- Use exterior-only glaze or burnish/oil finish for prototypes.

- Radius handling edges, but keep the labium clean and crisp.
- Fire sample bars with every clay batch.

Failure Modes To Watch

- Rounded labium edge: weak or unstable tone.
- Warped windway: airy or non-speaking note.
- Oversized initial holes: sharp pitch with no clean lowering path except wax.
- Thin wall near big bass holes: cracks during drying.
- Heavy glaze around holes: pitch shift and sluggish response.

photo-shotlist.md

Project artifact.

Gemshorn Photo Shot List

Generated: 2026-05-06

This shot list feeds the v4 build-log site and future capstone/print refreshes. Use real photos when prototypes exist; until then, the SVG drawings are placeholders.

Required Hero And Overview

Shot ID	Subject	Purpose	Status
HERO-001	Full family lineup, bass F through sopranino F	README and build-log hero	Needed
HERO-002	Historical horn blank beside ceramic body	Shows authentic vs production paths	Needed
HERO-003	Soprano C pilot in hand	Scale/ergonomics	Needed

Natural-Horn Build

Shot ID	Subject	Purpose	Status
HORN-001	Raw horn blanks, rejected and accepted examples	Blank-selection documentation	Needed
HORN-002	Cleaned horn cavity and closed tip	Shows boundary condition	Needed
HORN-003	Fitted removable block before windway cut	Fipple construction	Needed
HORN-004	Labium/window detail	Critical voicing reference	Needed
HORN-005	Final hole layout and tuning wax	Tuning method	Needed

Slip-Cast Ceramic Build

Shot ID	Subject	Purpose	Status
CER-001	Sealed master with split-line marks	Mold-making reference	Needed
CER-002	Two-part plaster mold open	Mold layout and registration	Needed
CER-003	Leather-hard cast with pilot holes	Greenware state	Needed
CER-004	Bisque body with temporary block	First playable test	Needed
CER-005	Finished glazed/burnished body	Final product	Needed

CNC-Routed Wood Build

Shot ID	Subject	Purpose	Status
CNC-001	Two wood halves on spoilboard with datum pins	Workholding proof	Needed
CNC-002	Routed cavity halves before glue-up	Shows internal geometry	Needed
CNC-003	Dry-fit leak test	Quality gate	Needed
CNC-004	Exterior shaping and fipple detail	CNC-to-handwork transition	Needed
CNC-005	Finished wood prototype next to ceramic pilot	Material comparison	Needed

Validation And Audio

Shot ID	Subject	Purpose	Status
VAL-001	Tuner reading for closed note	Evidence for `validation.csv`	Needed
VAL-002	Water-fill volume measurement	Helmholtz model correction	Needed
VAL-003	Hole diameter with pin gauge/calipers	Tuning traceability	Needed
VAL-004	Short audio/spectrogram capture	Build-log media	Needed

Naming Convention

Place future images under `images/`:

```
images/hero-family-lineup.jpg
images/horn-001-raw-blanks.jpg
images/cer-002-plaster-mold-open.jpg
images/cnc-002-routed-cavity-halves.jpg
images/val-001-closed-note-tuner.jpg
```

risks.md

Project artifact.

Risks

Generated: 2026-05-06

Acoustic

Acoustic: Helmholtz model overpredicts playable pitch

Symptom: The closed note or upper scale speaks flat/sharp even when the calculated cavity volume and tone-hole areas match the design sheet.

Mechanism: Gemshorn pitch depends on coupled fipple/window behavior, leakage, tone-hole chamfer, and effective neck length. The first-pass Helmholtz model is directionally useful but not a finished calibration.

Test: Build `GEM-SC-C5` first, measure fired water-fill volume, record the closed note and each opened-hole note at logged temperature, and compute cents error in `validation.csv`.

Mitigation: Update the design table with measured volume and empirical hole corrections before scaling to the full family. Keep holes flat/undersized before final tuning.

Severity: Medium.

Acoustic: CNC wood seam leakage weakens fundamental

Symptom: The CNC-routed wood version sounds airy, unstable, or low in volume compared with ceramic/horn.

Mechanism: Split-blank construction creates a long glue seam around the cavity. Even tiny leaks reduce acoustic conductance control and can change the fipple response.

Test: Before final shaping, plug the window and tone holes, gently pressurize the cavity by mouth or bulb, and brush soapy water along the seam.

Mitigation: Add a continuous shallow gasket groove, use a thin even glue line, and seal the interior before final glue-up if the finish is mouth-safe after cure.

Severity: Medium.

Structural

Structural: Ceramic cracks around large low-register holes

Symptom: Bass/tenor bodies crack during drying, bisque, drilling, or tuning around H6/T1 or the tuning vent.

Mechanism: Large holes interrupt a curved ceramic shell and create stress concentration. Wall thickness and drying gradients matter more on the larger family members.

Test: Make a clay coupon with the largest bass hole and wall target. Dry and bisque it before committing to the full bass mold.

Mitigation: Add local boss thickness around large holes, drill greenware undersize, slow dry under loose cover, and tune after bisque with gradual reaming.

Severity: High. Human decision required before cutting a bass mold if coupon fails.

Structural: Natural horn delaminates or cracks during cleaning

Symptom: A horn blank flakes, splits, or opens at the tip while being cleaned or drilled.

Mechanism: Horn is layered keratin. Heat, aggressive soaking, old dry material, or internal defects can delaminate the blank.

Test: Inspect with light through the wall, flex gently at the wide end, and clean one sacrificial blank before layout.

Mitigation: Buy multiple blanks, avoid boiling unless tested, keep the tip closed or permanently plug it, and reject thin/transparent wall sections near the voicing.

Severity: Medium.

Ergonomic

Ergonomic: Bass F body exceeds comfortable hand support

Symptom: The bass F is hard to hold, requires wrist extension, or pulls the mouthpiece angle downward.

Mechanism: The scaled bass body is long and wide enough that the instrument behaves like a small supported vessel rather than a hand flute.

Test: Make a cardboard or foam full-scale silhouette from `family-spec.csv` and check seated/lap, sling, and stand support positions.

Mitigation: Treat `GEM-SC-F3` as bench/lap/sling-supported, move holes for reach without assuming position strongly determines pitch, and add a support point to the design drawing.

Severity: Medium.

Ergonomic: Sopranino holes are too small for reliable playing

Symptom: The sopranino F plays inconsistently because fingertips cannot seal tiny holes precisely.

Mechanism: Scaling reduces tone-hole diameters and spacing. Small holes are tuning-sensitive and are easily rounded by finish or reaming.

Test: Make a flat fingerboard coupon with final hole diameters and ask the intended player to cover holes without looking.

Mitigation: Keep the sopranino as an optional family member, use a slightly larger body scale if needed, or omit chromatic ambitions on that size.

Severity: Low.

Supply

Supply: Ethical horn blanks are inconsistent

Symptom: Purchased horn blanks differ too much in curve, wall, odor, core condition, or closed-tip geometry.

Mechanism: Natural horn is not an industrial material. Blank descriptions rarely specify internal volume or fipple-friendly wide-end geometry.

Test: Order several candidate blanks and log length, wide OD, wall, tip condition, water volume, and rejection reason in `horn-blank-spec.csv` notes.

Mitigation: Buy extras, tune each horn as a one-off, and keep the ceramic/CNC paths as repeatable alternatives.

Severity: Medium.

Supply: Clay shrinkage data is missing or batch-specific

Symptom: Fired parts do not match master dimensions, and the family scale drifts.

Mechanism: Casting slip shrinkage varies by clay body, water content, mold age, drying, firing cone, and glaze schedule.

Test: Fire shrinkage bars with every clay batch and compare green, bone-dry, bisque, and glaze dimensions.

Mitigation: Update `gemshorn-design-table.xlsx` before master fabrication and keep masters editable until the clay body is locked.

Severity: High. Human decision required before production mold set if shrinkage is unmeasured.

Fit/Finish

Fit/Finish: Glaze or finish rounds the labium edge

Symptom: The instrument becomes breathy, slow to speak, or loses pitch stability after finishing.

Mechanism: The labium and windway need crisp, controlled edges. Glaze, oil, wax buildup, or sanding dust can change the edge and windway height.

Test: Photograph and measure the window before and after finish. Test-blow with the same temporary block before and after finishing.

Mitigation: Mask windway, labium, hole interiors, and block seats. Use exterior-only finish near the voicing and retouch the labium after firing/finish only with controlled tools.

Severity: High. Human decision required if the finish system cannot keep the voicing clean.

Fit/Finish: Wood body glue line is visible or uncomfortable

Symptom: The routed wood body has a visible seam, rough mouth feel, or finish witness line along the body.

Mechanism: Split-body construction exposes a longitudinal seam and can leave mismatched grain or finish absorption.

Test: Make one finish coupon with the planned glue, seal coat, and final finish; inspect under raking light and touch around the mouth area.

Mitigation: Choose straight-grained matched halves, place seam away from lips/fingers where possible, scrape flush before finish, and use a mouth-safe topcoat.

Severity: Low.

sources.md

Project artifact.

Authenticity Notes

Generated: 2026-05-02

What Counts As Authentic Here

For this packet, "authentic gemshorn" means a historically informed instrument rather than a museum-certified copy. The available evidence supports these constraints:

- The body is a short curved horn form, traditionally animal horn and plausibly ceramic in at least one late medieval find.
- It is blown from the wide/open end, which is closed by a fipple/block and voicing window.
- The pointed end stays closed, so the instrument behaves as a stopped inverted cone or vessel flute.
- The range is limited because it does not overblow like a recorder.
- The safest historical hole count is small. Four-hole evidence is stronger than the modern seven-front-hole consort layout.

What Is Modern In This Packet

The slip-cast consort family uses modern reconstruction logic:

- Repeatable molded ceramic bodies rather than individual animal horns.
- Seven front holes plus a back thumb/tuning vent for ensemble usability.
- Equal-temperament C/F family targets.
- Spreadsheet-derived hole diameters and scaling.

That modern line is useful and buildable, but it should not be labeled as a direct 15th-century copy.

Historically Informed Build Choices

- Make the first historical pilot in G4 because modern measured reconstructions around that register provide a dimensional sanity check.
- Use a waxed or replaceable tuning ring/vent near the voicing area. Historical and modern sources describe tuning by partially covering a vent/ring or thumb hole.
- Keep a soft, low-wind sound. Overblown recorder behavior is not expected.
- Keep exterior decoration restrained: polished horn, burnished ceramic, leather wrap, or simple incised marks. Avoid shiny glaze near the voicing.

Source Notes

- [Horace Fitzpatrick, The Gemshorn: a Reconstruction](https://www.cambridge.org/core/journals/proceedings-of-the-royal-musical-association/article/gemshorn-a-reconstruction/0AD60F0FE7748111C8455FA2F4DD1CC3): Defines the gemshorn as a fipple instrument made from animal horn and calls out the stopped inverted cone bore. Used as the boundary-condition anchor.
- [Early Music Muse, The gemshorn: a short history](https://earlymusicmuse.com/gemshorn/): Summarizes the open-end fipple construction, four-hole historical evidence, and the 1455 clay gemshorn-like find. Used for authenticity notes.
- [Cincinnati Early Music, Medieval Gemshorn](https://cincinnatiearlymusic.com/gemshorn.html): Contrasts historical four-hole evidence with modern recorder-like reconstructions and consort families. Used to separate historical and modern lines.
- [Baltimore Recorders, Information about the Gemshorn](https://www.baltimorerecorders.org/gemshorns.html): Describes front holes, thumb hole, whistle/fipple, tuning ring, and the lack of an outlet other than whistle and fingerholes. Used for construction details.
- [Institut fuer Musikforschung Wuerzburg, Lo 29-31 Gemshoerner](https://www.musikwissenschaft.uni-wuerzburg.de/musikinstrumente/bestand/inventarliste/lo29-31-gemshoerner/): Gives measured modern HFK gemshorns in g, d1, and g1 and cautions that modern gemshorns are reconstruction products. Used as dimensional sanity check.

Provenance Warning

No file in this packet claims to reproduce an extant original gemshorn exactly. The historical record is sparse; this packet gives a traceable and tunable workshop path with explicit assumptions.

tuning-and-fingering.md

Project artifact.

Tuning And Fingering

Modern Slip-Cast Consort

Hole names run from the distal/lower front hand toward the mouth:

```
T1 on back behind H7

Mouth / fipple
H7 H6 H5 H4 H3 H2 H1
Tip closed
```

First-pass diatonic opening sequence:

Sounding note above tonic	Fingering idea
0 semitones	all holes closed
+2	open H1
+4	open H1-H2
+5	open H1-H3
+7	open H1-H4
+9	open H1-H5
+11	open H1-H6
+12	open H1-H7
+14	open H1-H7 plus T1

Chromatic notes are by half-holing, cross-fingering, removable wax, or a rotating tuning ring. Treat the chart as a tuning scaffold, not a final published fingering chart.

Historical Archetype

Use the `hole-schedule-historical.csv` pilot. Build it with waxable holes and tune by ear/tuner:

Sounding note above tonic	Fingering idea
0 semitones	all holes closed
+2	open F1
+4	open F1-F2
+5	open F1-F3, optional
+7	open F1-F4
+12	open front holes plus T1/back vent

The strict historical version should be documented as "limited range, four-hole evidence, exact pitch unknown." The practical consort should be documented as "modern reconstruction."

Tuning Order

1. Tune closed tonic by cavity volume, window area, and any tuning vent.

2. Tune H1, then H2, working upward.

3. Tune the octave hole after the lower scale speaks reliably.

4. Tune T1 last. It doubles as high note, thumb vent, or pitch control.

5. Leave small holes slightly flat before glaze fire, then do final opening after glaze if needed.

Cents Formula

```
cents = 1200 * log2(measured_frequency / target_frequency)
```

Keep final production tolerance to +/-15 cents for first playable prototypes and tighten to +/-8 cents after shrinkage and voicing data are stable.

validation-report.md

Project artifact.

Validation Report

Generated: 2026-05-06

Verifier command:

```
python3 /mnt/c/Users/Tony/.codex/skills/instrument-maker-v4/scripts/validate_packet.py . --fix --json
```

Result: clean. No pass-1 findings and no fixes required.

Clean Checks

- Tier 1 packet files are present: design, BOM, sourcing, cut list, validation, assembly manual, RFQ, drawing brief, visual BOM brief, Wolfram starter, risks, capstone deck, print packet, manifest, and README.
- Capstone deck has 15 slides, above the v4 verifier threshold.
- README is no longer a minimal scaffold and references an existing hero drawing.
- Referenced drawing files exist.
- `print-packet.pdf` exists.
- `capstone-deck.pptx` exists.
- CSVs load successfully.
- SVG drawings parse as XML.
- `gemshorn-design-table.xlsx` passes zip integrity checks.

Fixed In v4 Verifier Loop

None. The final verifier pass reported no findings before applying fixes.

Escalated

None from the packet verifier.

Remaining Human Measurements

These are not verifier failures; they are shop measurements needed after first prototypes:

- Clay shrinkage by actual slip batch and firing schedule.
- Fired ceramic water-fill volume.
- Natural-horn blank measured volume and final hole diameters.
- CNC wood split-body leak test result.
- Measured Hz and cents error for each prototype.